

A wide-angle photograph of the Burj Khalifa in Dubai, United Arab Emirates, taken at dusk. The tower is the central focus, its spire reaching into a clear, light blue sky. The building's facade is illuminated with warm, golden lights, and its reflection is visible in the water below. The surrounding city skyline is also visible, with various other skyscrapers and buildings lit up. The overall scene is a vibrant display of modern architecture and urban development.

Burj Khalifa

Carolina Berkheimer-Lubeck, Michael Clariday, Sarah Lawley, Chloe Mengers, Cristina Robalino

Tower's spire visible from
60 miles
away

project cost about
US **\$4.1bn**

8 million
cubic feet of
concrete used in construction

22 million
man hours
spent building structure

26,000
glass panels for the
exterior cladding

10°C
cooler at
the top
than the
bottom

160 room
hotel
designed by
Georgio Armani

3,000
stairs from bottom to top

US **\$217m**
cost of the exterior
Dubai Fountain

part of entire new
Downtown Dubai
US **\$20bn**

Burj Dubai

Tallest building
in the world

on the 158th floor,
the world's tallest
mosque

on the 143rd floor,
the world's highest
nightclub

308.8m
taller than
Taipei 101
(world's next
tallest building)

from the 124th floor public observation deck,
you can see **IRAN**

height of the
Residential suites

on the 76th floor,
the world's highest
swimming pool

height of the
Armani hotel

www.menainfra.com

MENA
INFRASTRUCTURE

Content

Overview

Main Structural System

Foundation

Lateral Resisting System

Loads

Architect –

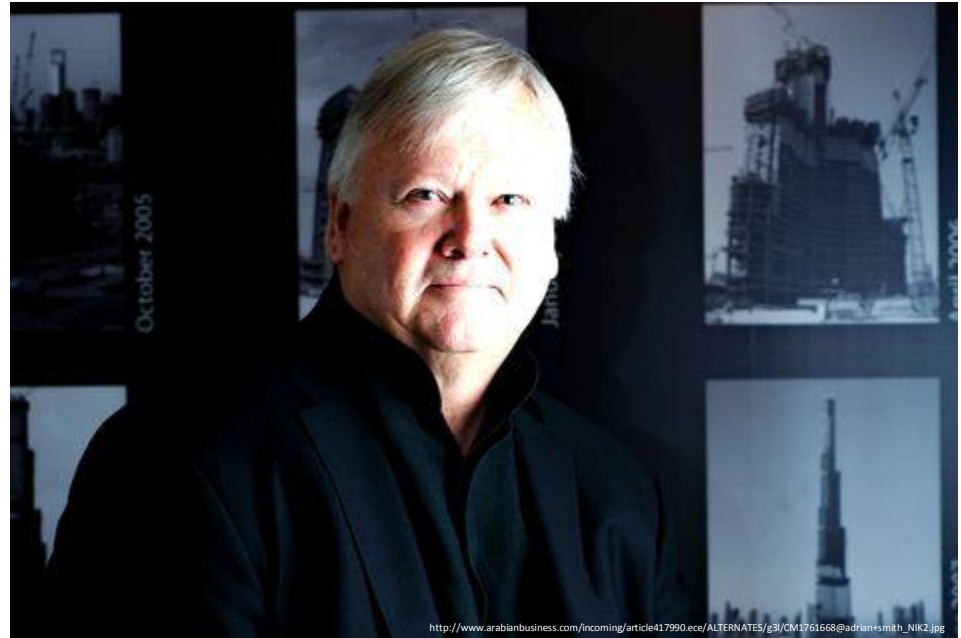
Adrian Smith at SOM

Structural Engineer –

Bill Baker at SOM

Developer –

Emaar Properties



“Test the limits of reinforced concrete”



"A tall tower would not only give the master plan an identity, but would also become the identity for EMAAR and the Country of Dubai"

- Adrian Smith

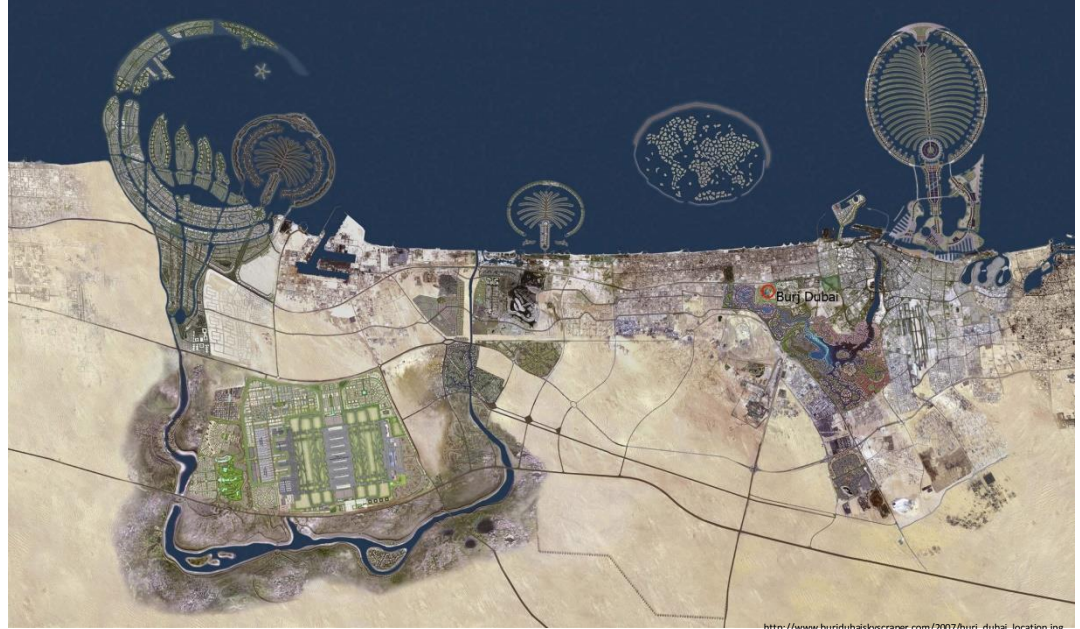
Overview Structural System Foundation Lateral Resisting System Loads

Location – 1 Sheikh
Mohammed bin Rashid
Boulevard, Dubai, United
Arab Emirates

Construction Started –
January 6th, 2004

Completed – December 31st,
2009

Cost - \$1.5 Billion in USD



http://www.burjdubaiskyscraper.com/2007/burj_dubai_location.jpg



http://www.burjdubaiskyscraper.com/2007/construction_timeline_05_24-11_23.jpg

Architectural Height –
828 m (2717 ft)

Tip Height – 829.8 m
(2722 ft)

Top Floor – 584.5 m
(1918 ft) (Level 154)

Observatory – 555.7 m
(1823 ft) (Level 148)



<http://balichws.com/wp-content/uploads/2016/03/Burj-Khalifa-Light-Show-1280x800.jpg>

Overview Structural System Foundation Lateral Resisting System Loads

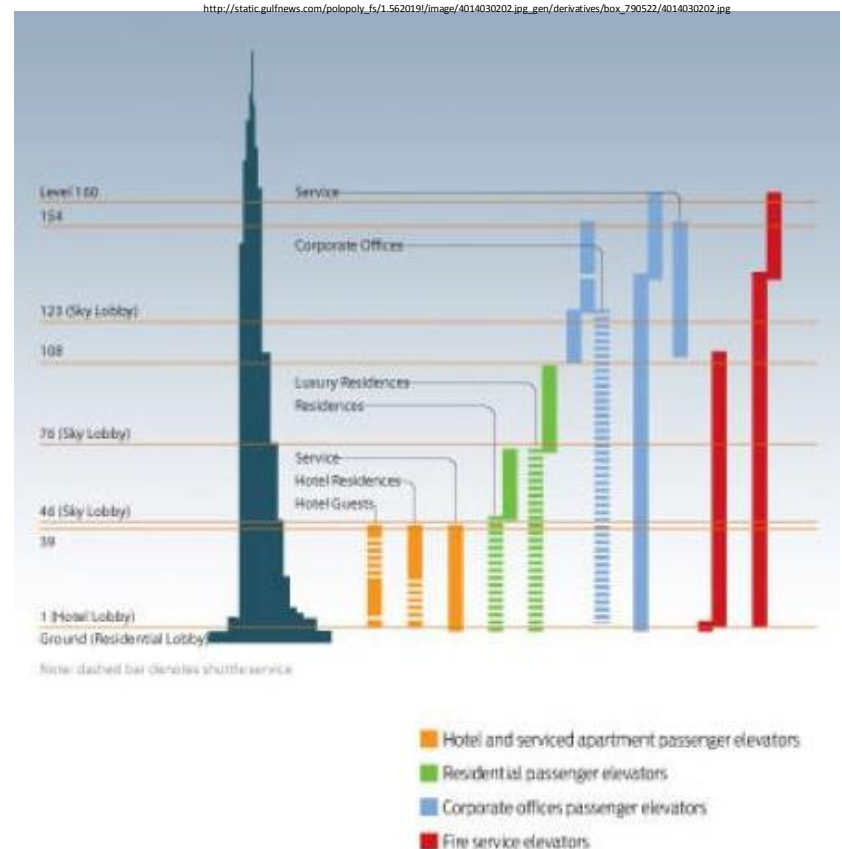
Program - hotels, commercial space, entertainment space, offices, and residential spaces

Material – Glass, steel, aluminum, and reinforced concrete

Floor Count – 163 above ground, 154 usable floors plus 9 maintenance levels, 46 spire levels, 2 below ground parking levels

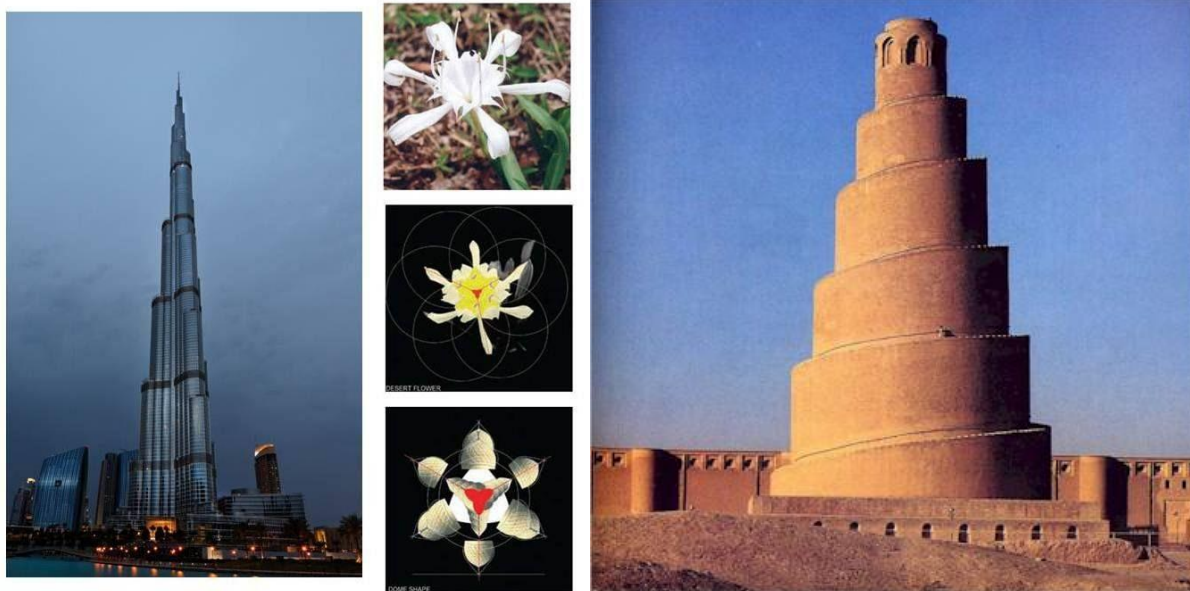
Floor Area – 309,473 m² (3,331,100 ft²)

Elevators – 57 (55 single deck and 2 double deck) by Otis Elevator Company



Tower Form and Concept

The initial concept for the design of the Burj Khalifa originated from the **hymenocallis flower**, and **spiral minarets** that are an important part of Islamic architecture.



Tower Form and Concept

The form was designed based on how to address the effect of the wind on the structure.

The wind force increases as the height of the building increases

How did they solve this?

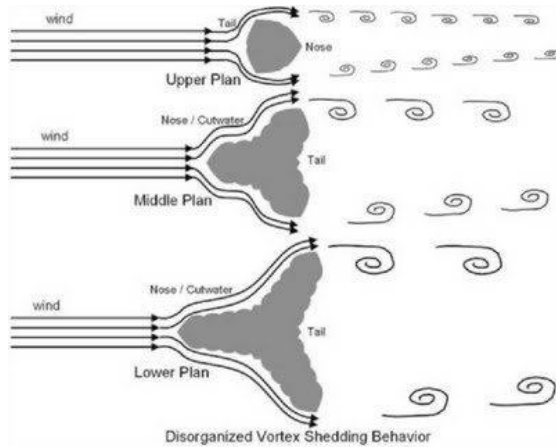
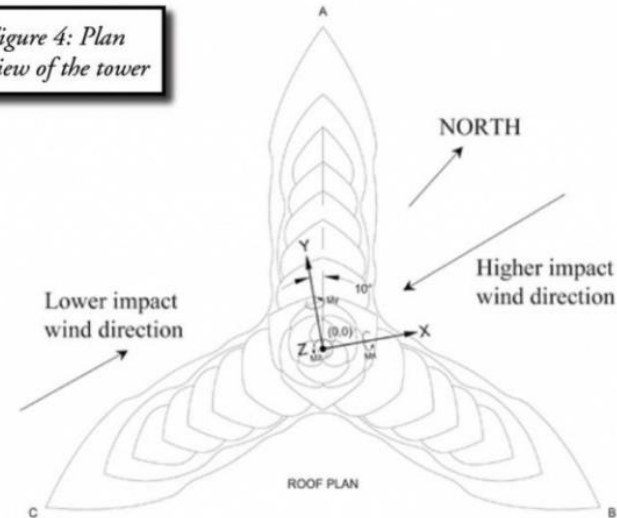


Figure 13. Wind behavior

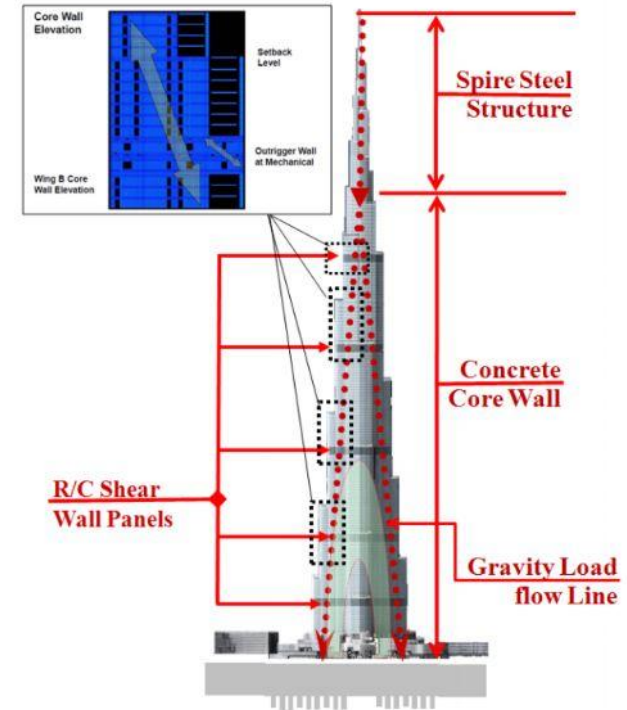
Figure 4: Plan view of the tower



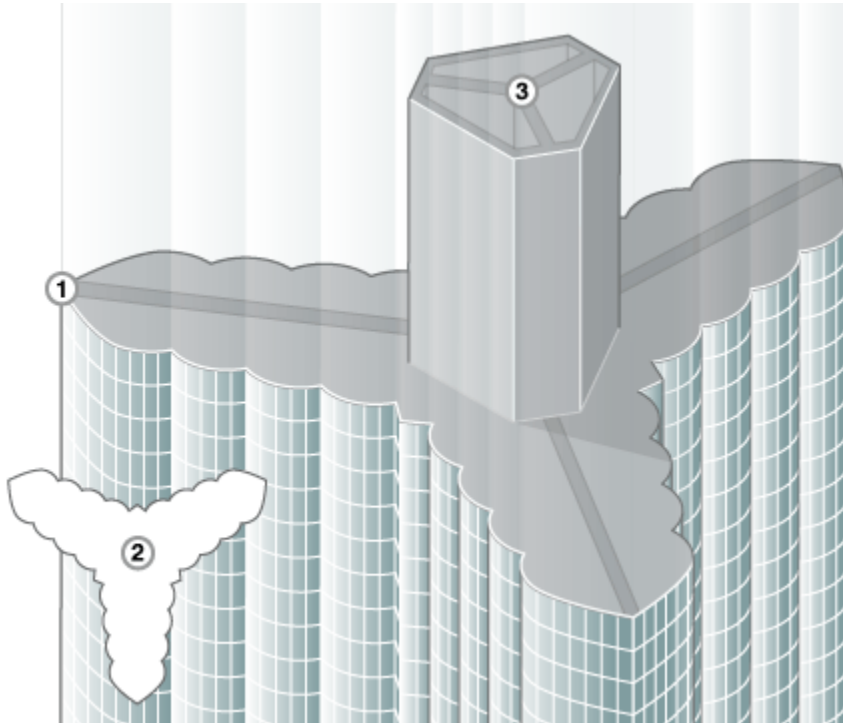
Key Objectives

Design a system that is strong, stiff, and effective

To achieve this, they used the latest technology in materials available locally, latest innovation in analysis, design, materials, and construction methods.



Main Elements

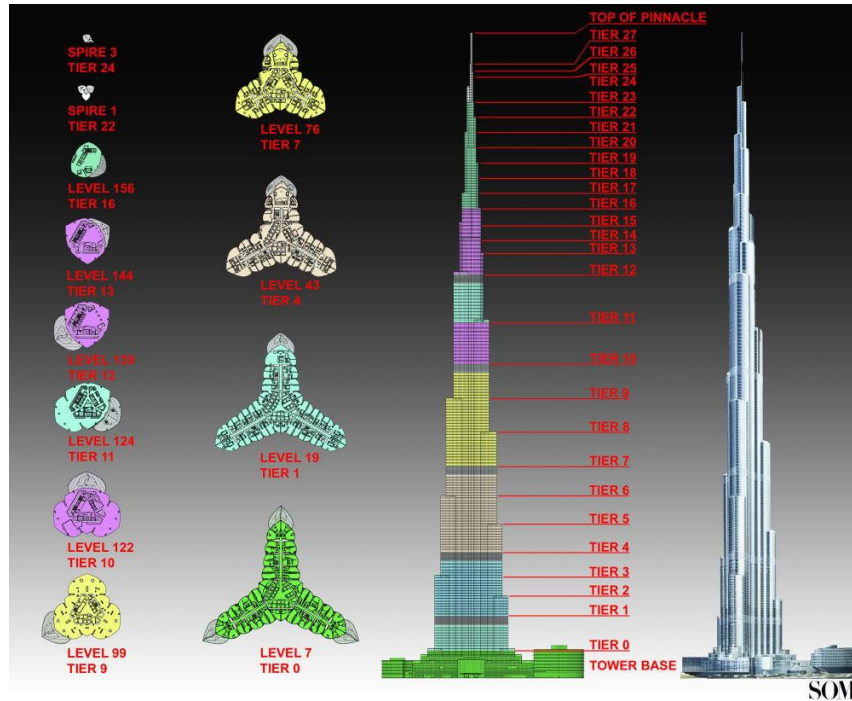


The main structural system for this megastructure has been named a **"buttressed core"**. This new structural system allows for a dramatic increase in height.

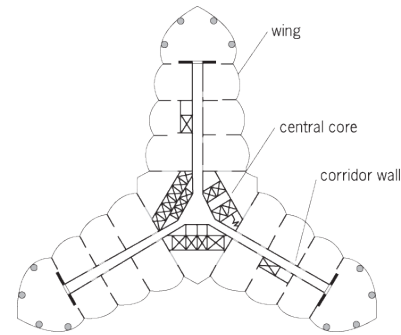
The structural system consists of a three-winged structure anchored to a strong hexagonal central core, each wing is buttressed to the other to provide a highly stable system.

The central core provides the torsional resistance for the building, while the wings provide the shear resistance and increased moment of inertia

Main Elements



The three wings attached to the central core have four bays each, forming a Y-shaped plan. At every seventh level one outer bay is removed from one of the wings as the tower spirals up. The Y-shaped plan is ideal for residential and hotel usage, with the wings allowing maximum outward views and inward natural light.

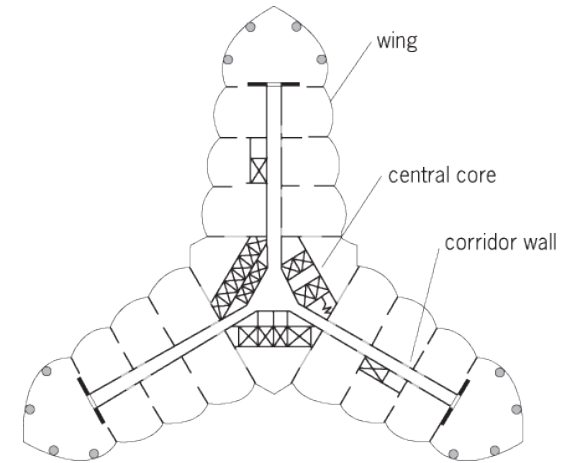


Main Elements



Within the wings, corridor walls extend from the hexagonal core to almost the end of the wing and end on thick walls named hammerhead walls.

With the addition of flat plate floor construction and perimeter columns, the entire structural system of the Burj Khalifa acts like a single unit creating a tower that acts as one giant concrete beam cantilevering from the ground.

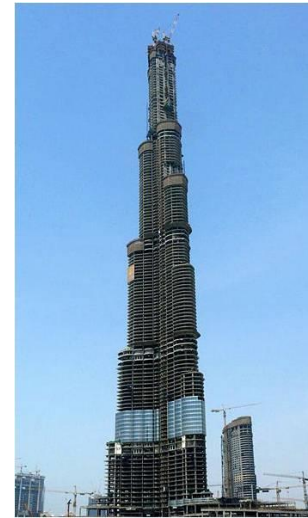


Structural Materials & Facade

High performance concrete with high compressive strength

Structural steel with a diagonally braced lateral system for the spire

Special exterior cladding specifically designed to withstand the temperature changes and wind forces



Existing Conditions

Geotechnical investigation: by *Grahame Bunce* of Hyder Consulting, the geotechnical engineer for the project and *Harry G. Poulos* of Coffey Geotechnics who is a technical reviewer for the geological aspects

Geotechnical investigations used various techniques using boreholes (roughly 30 boreholes were drilled).

Located on the Arabian Gulf = marine sediments

Low-lying land where sea level has changed throughout the periods

Located on the edge of the Arabian plate and close to Iranian Fold Belt = seismic area



<http://www.oobject.com/construction-of-the-burj-dubai/burj-dubai-foundation/6900/>

Reinforced concrete raft that is 3.7 meter thick and made of High Performance Self Compacting Concrete

Cathodic Protection System that creates a barrier to keep “eroding chemicals” from ground water affecting the concrete

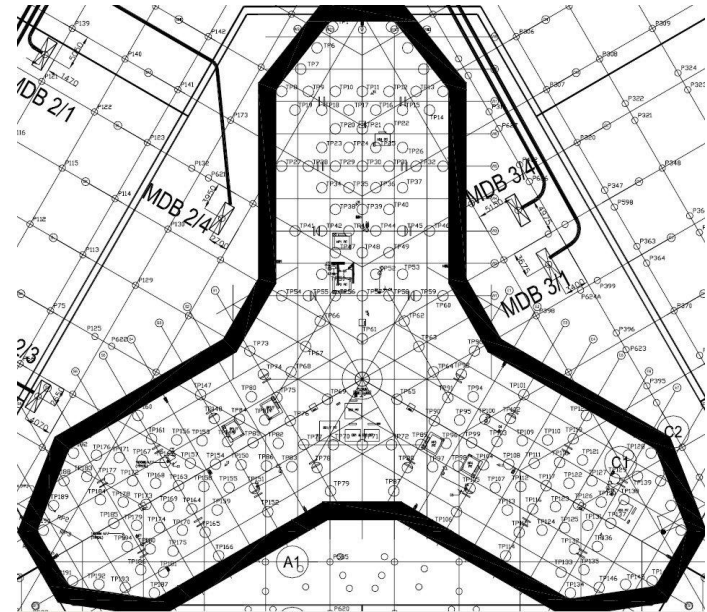
Low permeability and high density = greater protection from ground water

12500 cubic meters poured in four pours for raft foundation

192 piles that are 47 meters in length.

1.5 meters in diameter (piles) and placed 3.75 meters apart from center-to-center

Structural Overview



<http://www.slideshare.net/astoria104/foundation-design-of-burj-khalifa>

Structural Overview

Reinforced concrete raft that is 3.7 meter thick and made of High Performance Self Compacting Concrete

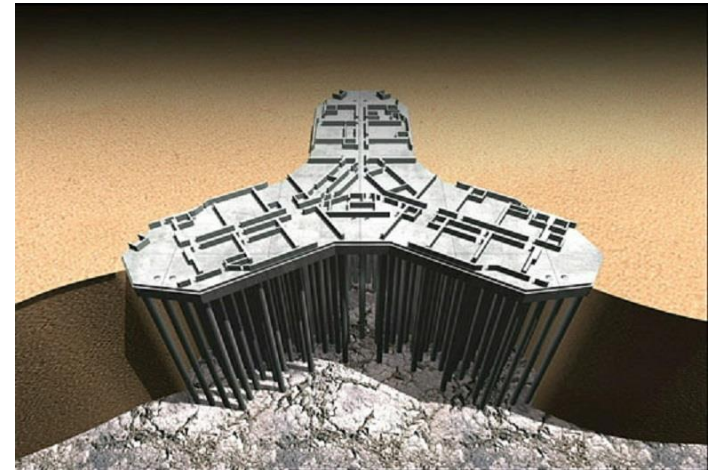
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Structural Overview



Allows entrance from on-grade to the first set of levels

Varying heights and entrances, allows for a transition between the three sides of the building

Entrances have glazing and allow for separate points of entry for the various levels including corporate levels, the Armani Hotel, and the private personal residences within the building.



Structure of Podium: suspended cable net
The winding tiers begin from this staggered base.

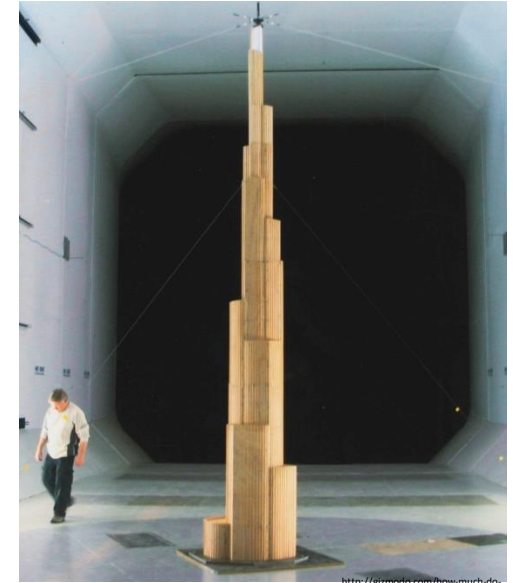
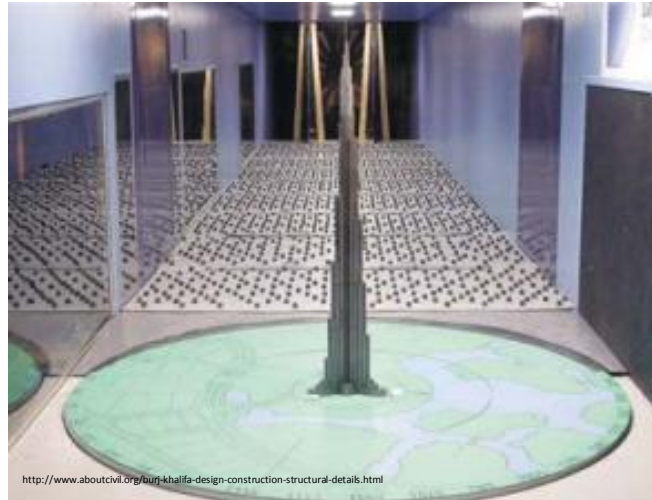
Wind Engineering

Determining the affect of
wind on a building

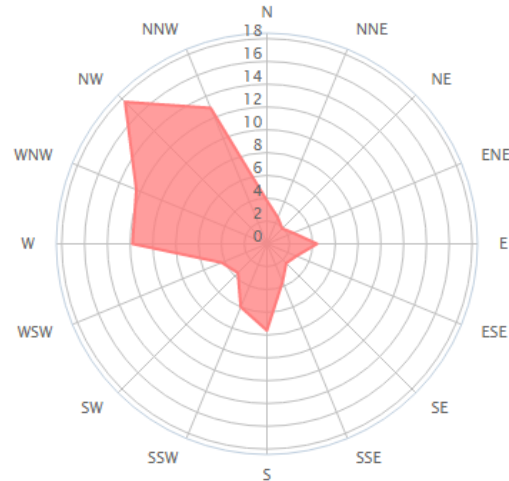
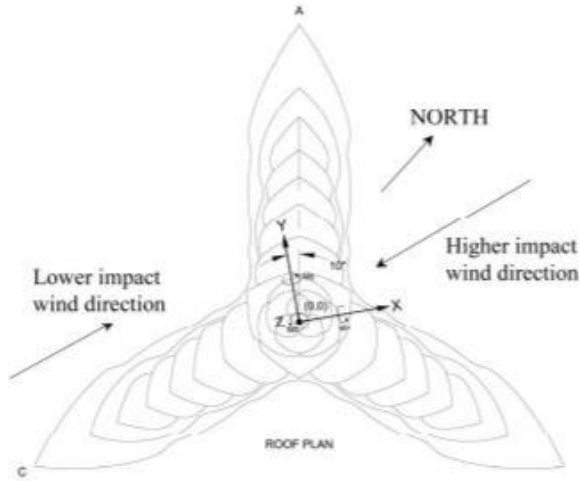
Over 40 studies done in
wind tunnel

Design Models

Construction Models



Building Orientation



Y-shape helps control wind pressure

6 wind directions

Building responds better to major wind impact on wings

Heavy wind directions in Dubai: Northwest and South

Core and Spire

Hexagonal Concrete Core

Resists torsion

Walls up to 1300mm (50inches) thick



700 feet tall

Over 400 tons of structural steel

Structural pipes up to 1200 mm (47 inches) in diameter

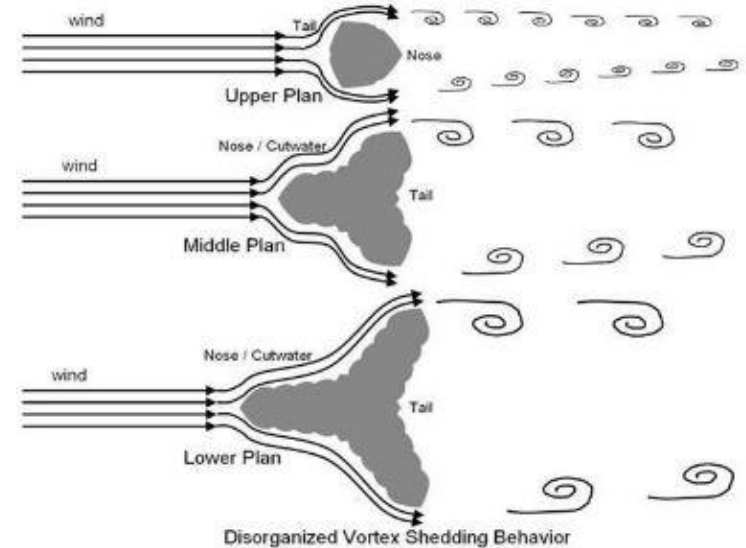




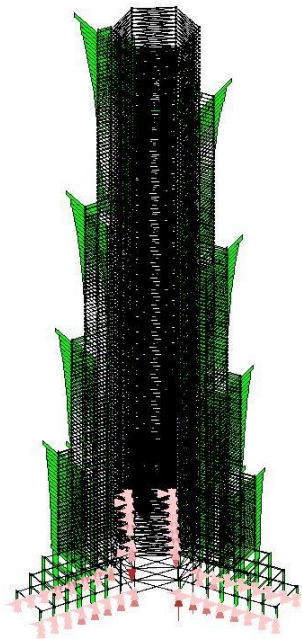
'Confuses the wind' to prevent
vortex shedding

Rotating Steps Backward

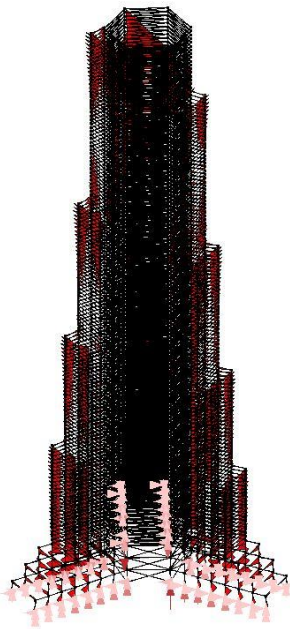
Wind Design



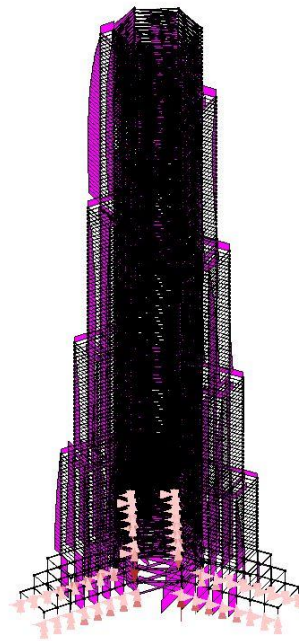
Multiframe



Shear Diagram



Moment Diagram



Axial Force Diagram

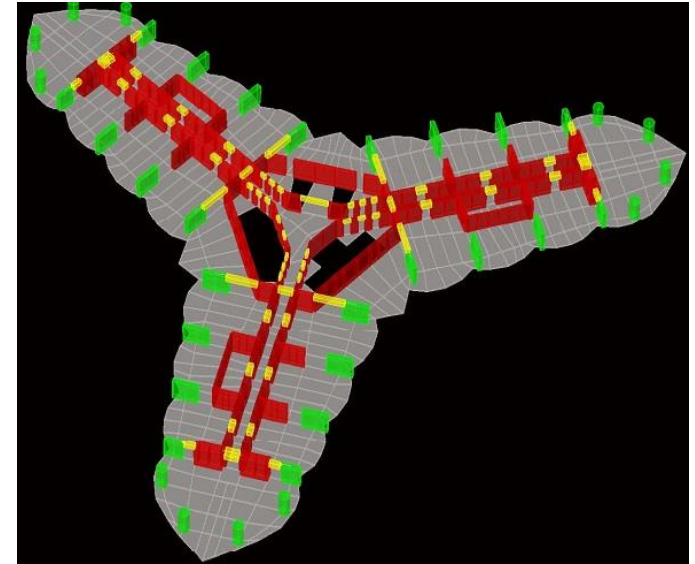
Seismic Design

Dubai is classified as a UBC97 Zone 2 seismic region with a seismic zone factor of $Z=0.15$ and the soil profile is S_c

It is a 'bundled tube with various heights combined in a three wing shape.'

Does not govern the design of the reinforced concrete structure

Was recognized in the design of the reinforced concrete Podium buildings and the spire composed of structural steel



<http://www.aboutcivil.org/images/Floor-analysis-model-of-bur-dubai.jpg>

Loading

Over 58,900 cubic yards of concrete was used to construct the concrete and steel foundation

The concrete core walls used 80 MPa of concrete from ground level to a vertical height of 601 meters

The perimeter columns are sized to match the same stress on the interior corridor walls

Outriggers at the mechanical floors allow the columns to participate in lateral load resistance

All of the vertical concrete is utilized to support both gravity and lateral loads



<https://i.ytimg.com/vi/KWBAtgicgZY/maredefault.jpg>

Key Design

Part of the load in a reinforced concrete column is transferred from the concrete to rebar

This reduces creep

Rebar in the columns and walls at Level 135 support about 15% of the load and the concrete supports 85%

After 30 years, rebar will support 30% and concrete supports 80%

The force in rebar increases as the steel ration is increased and/or as the total load decreases.

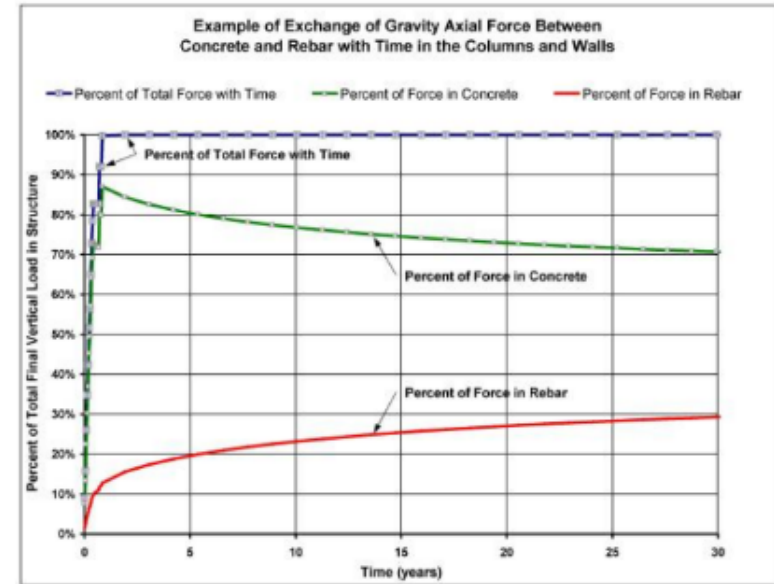


Figure 20 – Exchange of Gravity Axial Force between Concrete and Rebar vs. Time

Tower's spire visible from
60 miles
away

project cost about
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www.menainfra.com

MENA
INFRASTRUCTURE

In Conclusion

Latest modern Marvel

Tallest Man-made structure in the world

Race for the next Tallest building is on





Sources

~Any photos not marked came from Google

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